ASCEND TECH

AI Engineer Intern Technical Assessment

Overview

Develop an AI-powered application that allows users to virtually try on different eyeglasses using a live webcam.

The application should provide a realistic experience with accurate glasses positioning, smooth facial tracking, and minimal processing delay.

Choose One Development Approach

Option A: Build from Scratch

Develop the core virtual try-on functionality using computer vision and AI techniques.

You may use OpenCV, MediaPipe, PyTorch, TensorFlow, or other open-source frameworks.

Key Requirements:

- Real-time face detection and facial landmark tracking.
- Accurate glasses placement based on facial geometry.
- Realistic rendering of eyeglasses.
- Smooth tracking during head movement.
- Optimized performance with minimal latency.

Option B: Build Using APIs or Existing Models

Develop a functional virtual glasses try-on application by integrating third-party APIs, SDKs, or pretrained AI models.

Key Requirements:

- Integrate a suitable virtual try-on API, SDK, or AI model.
- Build a simple, interactive user interface.
- Support live webcam input where the selected technology allows it.
- Enable users to switch between different glasses.
- Optimize responsiveness and visual quality.
- Explain the API/model selection, integration process, limitations, and estimated usage costs.

ASCEND TECH

Expected Features

1. Live webcam-based virtual glasses try-on.
2. At least three different glasses designs.
3. Accurate alignment with facial movements.
4. Realistic visual appearance.
5. Smooth performance with minimal delay.
6. Simple and user-friendly interface.

Performance Targets

- **FPS:** Aim for 24 FPS or higher.
- **Latency:** Target less than 100 ms end-to-end.
- **Accuracy:** Glasses should remain aligned during natural head movements.
- **Realism:** Frames should look natural in different lighting conditions.

These are target metrics, not strict pass/fail requirements. Candidates should report actual results and testing conditions.

Submission Requirements

- GitHub repository containing source code.
- Working prototype or demo link.
- 1–2-minute demonstration video.
- Brief explanation of the technical approach.
- Details of APIs, models, or frameworks used.
- Performance measurements and known limitations.

Use of AI Tools & Vibe Coding

You are free to use **AI coding assistants, vibe coding platforms, and generative AI tools** to develop your application.

However, please follow these guidelines:

1. **Understand Your Code:** You must understand the AI-generated code and be able to explain its functionality, architecture, and logic.
2. **AI Credits:** Clearly acknowledge all AI tools used during development (e.g., ChatGPT, Claude, Cursor, GitHub Copilot, etc.).



3. **Prompt Submission:** Submit the prompts you used during development alongside your project, preferably in a separate AI_PROMPTS.md file.
4. **Code Ownership:** You should be able to debug, modify, and improve your code independently.
5. **Technical Interview:** Be prepared to explain your implementation, demonstrate your application, and answer technical questions about your code.

Remember: We are not evaluating how much code you write manually. We are evaluating your problem-solving skills, technical understanding, creativity, and ability to build working solutions using modern AI tools.

AI assistance is encouraged, but understanding your implementation is mandatory.

********** Good Luck **********